

The Identity as the Sole Universal Central Element in Group Theory

Pearl Bipin Pulickal

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Abstract

This article provides a rigorous proof that the identity element, e , is the only element guaranteed to belong to the center, $Z(G)$, of an arbitrary group G . We begin by formally establishing that the center of any group is precisely the intersection of all its centralizers. Using this fact, we construct a proof demonstrating that for any non-identity element g , a group always exists in which g is not central. This result solidifies the identity's foundational role as the universal anchor of commutativity in group theory. The argument is illustrated with detailed analyses of the symmetric group S_3 and the dihedral group D_4 .

1 Introduction

In group theory, the concepts of the *center* and the *centralizer* are fundamental to understanding a group's internal commutative structure. The center, $Z(G)$, may be trivial, as in the case of many symmetric groups, or it may encompass the entire group, as in any abelian group. This raises a natural question: are there any elements that are guaranteed to be in the center of *every* group, regardless of its structure?

This paper demonstrates that the identity element, e , holds this property uniquely. We formalize this by proving two key results:

1. The center $Z(G)$ is the intersection of all centralizers $C_G(x)$ for $x \in G$.
2. The identity element e is the only element that belongs to $Z(G)$ for every group G .

This perspective not only reinforces the definition of the identity but also provides a deeper insight into the nature of commutativity itself.

2 Preliminaries: Center and Centralizer

We begin by stating the standard definitions that form the basis of our argument.

Definition 2.1 (Center of a Group). The *center* of a group G , denoted $Z(G)$, is the set of elements that commute with every element in G :

$$Z(G) := \{z \in G \mid zx = xz \text{ for all } x \in G\}.$$

It is a fundamental result that $Z(G)$ is always an abelian subgroup of G .

Definition 2.2 (Centralizer of an Element). The *centralizer* of an element $a \in G$, denoted $C_G(a)$, is the set of elements in G that commute with a :

$$C_G(a) := \{g \in G \mid ga = ag\}.$$

For any $a \in G$, $C_G(a)$ is a subgroup of G . The center can be seen as the set of elements whose centralizer is the entire group, i.e., $Z(G) = \{z \in G \mid C_G(z) = G\}$.

3 The Universal Centrality of the Identity

With the foundational definitions in place, we can now establish the main result. The first step is to formally connect the center to the collection of all centralizers.

Proposition 3.1 (The Center as an Intersection of Centralizers). *For any group G , the center is the intersection of all centralizers of elements in G .*

$$Z(G) = \bigcap_{a \in G} C_G(a).$$

Proof. Let $g \in Z(G)$. By definition, $gx = xg$ for all $x \in G$. This implies that for any specific element $a \in G$, g commutes with a . Therefore, $g \in C_G(a)$ for all $a \in G$, which means $g \in \bigcap_{a \in G} C_G(a)$.

Conversely, let $g \in \bigcap_{a \in G} C_G(a)$. This means g is an element of every centralizer. By definition, $g \in C_G(a)$ implies $ga = ag$. Since this holds for all $a \in G$, g commutes with every element of G . Therefore, by the definition of the center, $g \in Z(G)$. $\square$

This proposition provides the foundation for our main theorem. We know by definition that the identity element e commutes with all elements ($ex = x = xe$), so $e \in Z(G)$ for any group G . The following theorem proves that no other element shares this universal property.

Theorem 3.2 (The Identity as the Sole Universal Central Element). *The identity element e is the only element that is guaranteed to belong to the center $Z(G)$ for every group G .*

Proof. We have already established that $e \in Z(G)$ for any group G .

To prove its uniqueness, we must show that for any non-identity element $g \neq e$, there exists at least one group H such that $g \notin Z(H)$.

Let G be any group containing a non-identity element g .

- **Case 1:** g is not in the center of G . In this case, we have found a group (namely G itself) where g is not central, and the proof is complete for this case.
- **Case 2:** g is in the center of G (i.e., $g \in Z(G)$). We must construct a new, larger group based on G in which g is no longer central.

Consider the symmetric group S_3 , which is non-abelian and has a trivial center, $Z(S_3) = \{e_{S_3}\}$. Let $\tau = (1\ 2)$ be a transposition in S_3 .

Now, construct the direct product group $H = G \times S_3$. The elements of H are ordered pairs (h, k) where $h \in G$ and $k \in S_3$. The group operation is component-wise: $(h_1, k_1)(h_2, k_2) = (h_1h_2, k_1k_2)$.

The element g from G corresponds to the element (g, e_{S_3}) in H . Let us test if this element is in the center of H . For an element to be in $Z(H)$, it must commute with all other elements in H .

Consider the element $(e_G, \tau) \in H$. We check for commutation:

$$(g, e_{S_3}) \cdot (e_G, \tau) = (g \cdot e_G, e_{S_3} \cdot \tau) = (g, \tau).$$

$$(e_G, \tau) \cdot (g, e_{S_3}) = (e_G \cdot g, \tau \cdot e_{S_3}) = (g, \tau).$$

This appears to commute. Let's re-evaluate the proof strategy. The element from S_3 must not commute with another element in S_3 . Let $\sigma = (1\ 2\ 3) \in S_3$. We know $\tau\sigma \neq \sigma\tau$.

Let's re-check the center of a direct product. The center of a direct product is the direct product of the centers: $Z(G \times S_3) = Z(G) \times Z(S_3)$. Since $Z(S_3) = \{e_{S_3}\}$, the center of H is $Z(H) = Z(G) \times \{e_{S_3}\}$.

The element in H corresponding to g is (g, e_{S_3}) . For this element to be in $Z(H)$, we must have $(g, e_{S_3}) \in Z(G) \times \{e_{S_3}\}$. This requires $g \in Z(G)$ and $e_{S_3} \in \{e_{S_3}\}$, which is true by our assumption for this case.

The proof needs a group construction where the element from G does not commute with an element from the other group. A semi-direct product is the standard tool.

Let's try a simpler, more direct argument.

Revised Proof: Let g be any element from any group such that $g \neq e$. We must show there exists a group where g is not central. By Cayley's Theorem, any group G is isomorphic to a subgroup of the symmetric group $S_{|G|}$ on the set of its elements. Let's call this embedding $\phi : G \rightarrow S_{|G|}$. The element g is mapped to a permutation $\phi(g)$.

If $|G| > 2$, the group $S_{|G|}$ has a trivial center: $Z(S_{|G|}) = \{id\}$. Since $g \neq e$, its corresponding permutation $\phi(g)$ is not the identity permutation. Thus, $\phi(g) \notin Z(S_{|G|})$. We have found a group $(S_{|G|})$ containing an isomorphic copy of g where it is not central.

If $|G| = 2$, then $G \cong \mathbb{Z}_2$, which is abelian, so its non-identity element g is central. We can embed G into S_3 (e.g., mapping g to (12)). Since $Z(S_3) = \{e\}$, the element (12) is not central in S_3 .

Therefore, for any non-identity element g , we can always find a group in which it (or an isomorphic copy) is not central. This completes the proof that e is the only universal central element.

□

4 Illustrative Examples

We examine two canonical non-abelian groups to illustrate these principles.

4.1 The Symmetric Group S_3 (A Trivial Center)

The group $S_3 = \{e, (12), (13), (23), (123), (132)\}$ is the smallest non-abelian group. Its center is trivial, $Z(S_3) = \{e\}$. We can verify this by computing the intersection of its centralizers. It suffices to check one element from each conjugacy class. The conjugacy classes are $\{e\}$, $\{(12), (13), (23)\}$, and $\{(123), (132)\}$.

Element Representative a	Centralizer $C_{S_3}(a)$
e	S_3
(12)	$\{e, (12)\}$
(123)	$\{e, (123), (132)\}$

The intersection of these subgroups is:

$$Z(S_3) = C_{S_3}(e) \cap C_{S_3}(12) \cap C_{S_3}(123) = S_3 \cap \{e, (12)\} \cap \{e, (123), (132)\} = \{e\}.$$

This confirms that only the identity is central in S_3 .

4.2 The Dihedral Group D_4 (A Non-Trivial Center)

The dihedral group D_4 represents the symmetries of a square. It is generated by a rotation r (by 90°) and a flip s , with relations $r^4 = s^2 = e$ and $srs = r^{-1}$. The group has 8 elements. Its center is non-trivial: $Z(D_4) = \{e, r^2\}$, where r^2 is the 180° rotation. The conjugacy classes are $\{e\}$, $\{r^2\}$, $\{r, r^3\}$, $\{s, r^2s\}$, and $\{rs, r^3s\}$.

Element	Representative a	Centralizer $C_{D_4}(a)$
e		D_4
r^2		D_4
r		$\{e, r, r^2, r^3\} = \langle r \rangle$
s		$\{e, s, r^2, r^2s\}$
rs		$\{e, rs, r^2, r^3s\}$

The intersection of all centralizers is:

$$\bigcap_{a \in D_4} C_{D_4}(a) = C_{D_4}(r) \cap C_{D_4}(s) = \{e, r, r^2, r^3\} \cap \{e, s, r^2, r^2s\} = \{e, r^2\}.$$

This matches $Z(D_4)$ exactly, illustrating Proposition 3.1. While r^2 is central in D_4 , it is not universally central (e.g., it is not an element of S_3). Only e is guaranteed to be in the center of both D_4 and S_3 , and indeed, of all groups.

5 The Limiting Case of Abelian Groups

For completeness, consider an abelian group G . By definition, every element commutes with every other element. This has two immediate consequences for our framework:

1. The centralizer of any element $a \in G$ is the entire group: $C_G(a) = G$.
2. The center is therefore the entire group: $Z(G) = \bigcap_{a \in G} C_G(a) = G$.

This case of maximal centrality is not an exception but an endpoint on the spectrum of commutativity. It is perfectly consistent with our findings, and the identity e is, of course, a member of $Z(G)$.

6 Conclusion

By establishing that the center of a group is the intersection of all its centralizers, we gain a powerful lens through which to view commutativity. This viewpoint allows for a rigorous proof that the identity element, e , is unique in its universal membership in the center of any group. While other elements may be central in specific groups, only the identity is an axiomatically central element across all of group theory. It is the absolute and unchanging foundation upon which the commutative structure of any group is built.